RELATIVE DIMENSION BOUNDS APPEAR AT A FINITE STAGE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2085-2126, following the statement at 2072-2083.

Editorial note. No omissions; the etale-localization equivalence of relative dimension definitions is justified in the text.

Allowed standard prerequisites (editorial).

- Relative dimension of morphisms of spaces. Morphisms of Spaces, Definition spaces-morphisms-definition-relative-dimension and spaces-morphisms-definition-dimension-fibre (cross-chapter, cited at ~2078-2079 and ~2119-2122). The equivalence X_i -> Y_i iff U_i -> V_i, via etale localization.
- Relative dimension for schemes. Morphisms, Definition morphisms-definition-relative-dimension-d (cross-chapter, cited at ~2117-2118). The corresponding notion for the affine schemes U_i -> V_i.
- Dimension of limits of schemes. Limits, Lemma limits-lemma-limit-dimension (cross-chapter, cited at ~2124-2125). The scheme-level engine: U -> V has relative dimension <= d iff U_i -> V_i does for some i.

Source reference. Original reference 05M5: focused-seeds/geometry/S9-spaces-limits-eventually-relative-dimension/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02NJ: focused-seeds/geometry/S9-spaces-limits-eventually-relative-dimension/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084W: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1666–1666. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NM: focused-seeds/geometry/S9-spaces-limits-eventually-relative-dimension/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 06LR: focused-seeds/geometry/S9-spaces-limits-eventually-relative-dimension/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Assumptions and notation as in Situation 084W. Let $d \geq 0$. If*

(1) *f has relative dimension $\leq d$ (Morphisms of Spaces, Definition 06LR), and*
(2) *f_0 is locally of finite type,*

then there exists an i such that f_i has relative dimension $\leq d$.

Proof. Choose an affine scheme V_0 and a surjective étale morphism $V_0 \to Y_0$. Choose an affine scheme U_0 and a surjective étale morphism $U_0 \to V_0 \times_{Y_0} X_0$. Diagram

$$\begin{array}{ccc} U_0 & \longrightarrow & V_0 \\ \downarrow & & \downarrow \\ X_0 & \longrightarrow & Y_0 \end{array}$$

The vertical arrows are surjective and étale by construction. We can base change this diagram to B_i or B to get

$$\begin{array}{ccc} U_i & \longrightarrow & V_i \\ \downarrow & & \downarrow \\ X_i & \longrightarrow & Y_i \end{array} \quad \text{and} \quad \begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

Note that U_i, V_i, U, V are affine schemes, the vertical morphisms are surjective étale, and the limit of the morphisms $U_i \to V_i$ is $U \to V$. In this situation $X_i \to Y_i$ has relative dimension $\leq d$ if and only if $U_i \to V_i$ has relative dimension $\leq d$ (as defined in Morphisms, Definition 02NJ). To see the equivalence, use that the definition for morphisms of algebraic spaces involves Morphisms of Spaces, Definition 04NM which uses étale localization. The same is true for $X \to Y$ and $U \to V$. Since f_0 is locally of finite type, so is the morphism $U_0 \to V_0$. Hence the lemma follows from the more general Limits, Lemma 05M5. □